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Next item →

Instructions

1. Which of the following properties is essential for a function to be considered a linear mapping? 1 / 1 point

- (•) Both A and B
- ( ) Preservation of vector addition
- ( ) None of the above
- ( ) Preservation of scalar multiplication

Nice work

Correct! Both the preservation of vector addition and scalar multiplication are essential properties for a function to be considered a linear mapping.

2. Given a linear mapping $\Phi : R^2 \to R^2$ defined by $\Phi(x, y) = (2x + y, x-y)$, what would be the matrix representation A of Φ? 1 / 1 point

- ( ) $\begin{bmatrix} 1 & 1 \\ 2 & -1 \end{bmatrix}$
- (•) $\begin{bmatrix} 2 & 1 \\ 1 & -1 \end{bmatrix}$
- ( ) $\begin{bmatrix} 2 & 1 \\ 1 & 0 \end{bmatrix}$
- ( ) $\begin{bmatrix} 2 & 0 \\ 1 & -1 \end{bmatrix}$

Nice work

Correct! This matrix represents the linear mapping $\Phi(x, y) = (2x + y, x-y)$.

3. Why are linear mappings important in machine learning? 1 / 1 point

- ( ) They solve differential equations.
- (•) They allow for the transformation and manipulation of data, including dimensionality reduction and feature extraction.
- ( ) They are only used in neural networks.
- ( ) They help in defining loss functions.

Nice work

Correct! Linear mappings are important in machine learning because they allow for the transformation and manipulation of data, including dimensionality reduction and feature extraction.

4. How can a linear mapping Φ:R^n→R^m be represented? 1 / 1 point

- ( ) As a function graph
- ( ) As a scalar
- (•) As an m×n matrix
- ( ) As a polynomial equation

Nice work

Correct! A linear mapping from R^n→R^m can be represented as an mxn matrix.

5. Changing the basis of a vector space affects the outcome of a linear transformation. 1 / 1 point

- (•) False
- ( ) True

Nice work

Correct! Changing the basis of a vector space does not affect the outcome of a linear transformation; it only changes its representation.

6. What is the purpose of changing bases in the context of linear mappings? 1 / 1 point

- (•) To simplify the representation of linear transformations
- ( ) To modify the size of a matrix
- ( ) To convert a matrix into a polynomial
- ( ) To add two matrices together

Nice work

Correct! Changing bases can simplify the representation of linear transformations, making them easier to work with.

7. What is a linear mapping (or linear transformation)? 1 / 1 point

- (•) A function that takes vectors from one vector space and maps them to vectors in another vector space while preserving vector addition and scalar multiplication.
- ( ) A method of solving systems of equations.
- ( ) A process of changing coordinates from one basis to another.
- ( ) A function that takes matrices from one space and maps them to another.

Nice work

Correct! A linear mapping takes vectors from one vector space and maps them to vectors in another vector space while preserving vector addition and scalar multiplication.

8. If a transformation Φ does not preserve scalar multiplication, it can still be considered a linear mapping. 1 / 1 point

- (•) false
- ( ) true

Nice work

Correct! A transformation must preserve scalar multiplication to be considered a linear mapping.

9. If Φ:V→W is a linear mapping, which of the following statements is correct? 1 / 1 point

- ( ) Φ(x+y)=Φ(x)+Φ(y) for all vectors x,y∈V
- ( ) Φ(ax)=aΦ(x) for any scalar aaa and vector x∈V
- (•) Both A and B
- ( ) Neither A nor B

Nice work

Correct! Both statements are true for a linear mapping: it preserves vector addition and scalar multiplication.